

FRAUD DETECTION ANALYSIS DASHBOARD

Power BI Project Documentation

Theme: Turning Data into Safer Transactions

1. Project Overview

The Fraud Detection Analysis Dashboard is an interactive Power BI dashboard designed to analyze transaction activity and identify important fraud patterns across transaction types, device types, locations, transaction amounts and time. The dashboard combines executive KPIs with supporting visuals on a single page.

2. Project Objectives

- Monitor total transaction volume.
- Separate genuine and fraudulent transactions.
- Measure the overall fraud rate.
- Track fraud trends over time.
- Identify transaction types with higher fraud activity.
- Analyze fraud by device type and location.
- Compare fraudulent amounts across transaction types.
- Present the findings in an executive-friendly dashboard.

3. Tools & Technologies

Power BI Desktop was used for dashboard development and visualization. Power Query was used for data preparation and transformation. DAX was used for analytical measures and KPIs. A date dimension supports time-based analysis.

4. Key Performance Indicators

KPI	Value	Purpose
Total Transactions	1,200	Overall transaction volume
Genuine Transactions	1,117	Transactions classified as genuine
Fraud Transactions	83	Transactions classified as fraudulent
Fraud Rate	6.92%	Share of transactions identified as fraudulent
Fraud Amount	149.15K	Total amount associated with fraudulent transactions

5. Dashboard Visuals

Visual	Purpose
Monthly Transaction & Fraud Trend	Shows transaction volume and fraud movement across Year Month.
Fraud Transactions by Transaction Type	Treemap comparing fraud transaction counts across Transfer, Debit, Credit, Payment and Withdrawal.
Fraud Transactions by Device Type	Donut chart showing fraud distribution across Mobile, Web and ATM.
Fraud Amount by Transaction Type	Pie chart comparing fraudulent amount across transaction types.
Top Locations by Fraud Transactions	Horizontal bar chart showing the leading locations by fraud transaction count.
Fraud Rate Gauge	Displays the fraud rate on a 0%–10% scale with a 5% reference target.

6. Key Dashboard Observations

- 83 fraudulent transactions are displayed out of 1,200 total transactions.
- The fraud rate is 6.92%, above the 5% reference target shown on the gauge.
- Mobile accounts for 44 fraud transactions (53.01%), Web for 31 (37.35%), and ATM for 8 (9.64%).
- Kolkata is the leading location in the displayed Top Locations chart, followed by Kochi, London, Bengaluru, Dubai and Pune.
- Transfer and Debit occupy the largest areas in the fraud transaction treemap.
- Transfer and Debit represent the largest portions of the fraud amount distribution.

7. Data Model & Analytical Structure

The dashboard uses a transaction fact table together with a date dimension. The date dimension supports fields such as Date, Month, Quarter, Year, Year Month and YearMonthSort. The transaction table provides analytical attributes including Location, DeviceType, TransactionType, Fraud, Fraud Amount and Fraud Transactions.

8. Dashboard Design

The final dashboard uses a single-page executive layout. Five KPI cards are positioned at the top for rapid monitoring. The remaining area contains a trend chart, treemap, donut chart, pie chart, location ranking and gauge. The KPI cards are visually highlighted with bordered panels.

9. Business Value

The dashboard helps fraud and risk teams quickly identify where fraudulent activity is concentrated, which transaction types and devices require attention, which locations show higher fraud counts, and whether the overall fraud rate is above the selected reference target.

10. Conclusion

This project demonstrates an end-to-end Power BI workflow covering data preparation, data modeling, DAX-based KPI creation, visualization and dashboard design. The final one-page dashboard brings the major fraud indicators together in a compact format suitable for portfolio presentation and interview discussion.